Statement of Work (SOW) for Coated Fabric Testing

Project Title: Tensile, Tear, Puncture Stiffness Testing at Hot, Ambient, and Cryo Conditions

Background: The overarching goal of the NASA's Artemis Suit Material (ASM) Project conducted by the Crew and Thermal System Division Advanced Suit Team, is to develop enabling technologies and potentially a turnkey solution for suit vendors for the outer shell fabric of a lunar suit environmental protection garment (EPG). The project asserts that no existing fabric currently meets all requirements necessitating the construction of a bespoke solution designed to withstand the harsh conditions of the lunar environment.

For this reason, the ASM Team is seeking the capability to strength test various prototype fabrics at lunar like temperature conditions to characterize their performance in order to down select to the best combination of options.

Objective: The Vendor will receive various test specimens, pre-cut to size, per specified ASTM standards. Then, measurements are to be made using the United Testing System (UTS) STN-100KN system at NASA specified temperature conditions.

Period of Performance: 6 months after ATP. The work will be performed on an "as needed" basis when the NASA ASM Team has specimens ready for testing.

Scope of Work:

1. **Check in samples:** inspect for damage, photograph, label and store in a secure environment.
2. **Specimen preparation:** Verify dimensions of specimens in their as received condition. Provide in table format:
    a. **Specimen ID**
    b. **Sample ID**
    c. **Sample Description**
    d. **Length (mm)**
    e. **Width (mm)**
    f. **Thickness (mm)**
3. **Specimen Conditioning:** Monitor temperature prior to test with thermocouple. Allow specimen to reach a stable temperature and maintain for 30 minutes prior to test. Remove thermocouple before test.
4. **Test Procedure:**
    a. Test type and target temperature will be specified by the NASA ASM Team prior to specimen shipment
    b. Test Types:
        i. ASTM D4032-08, "Standard Test Method for Stiffness of Fabric by the Circular Bend Method"

            1. Note: this procedure has been withdrawn by ASTM but is approved for use by the NASA ASM team
    ii. ASTM D2261-13, “Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)”
    iii. ASTM F1342/F1342M-05, “Standard Test Method for Protective Clothing Material Resistance to Puncture”
    iv. ASTM D5035-14, “Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)”
  c. Target Test Temperature:
    i. Ambient: 23.0 °C
    ii. Hot: 121.0 °C
    iii. Cryo: -196.0 °C
  d. Log force with load cell and log temperature with thermocouple input module.
  e. Compute average and maximum force required for each specimen type tested.
  f. Collect at minimum 100 data points
  g. Provide photographs and videos of test setup and test results deemed of interest.
  h. Provide final report with table of test results and photos.

Travel: No travel expenses are required/included in this TO.

Deliverables:

- Final report with table of test results and photos for each specimen and each test type.